Validation report

Hand-calculated benchmark for two alternatives (VRF vs water-cooled chiller), compared with GreenLCC India.

Result. Every one of the 14 comparisons between the independent Excel benchmark and the tool agrees. The largest difference is below one hundred-millionth of a rupee (floating-point noise). A second, closed-form method (interest factors) agrees with the year-by-year sum to within 1e-07 rupees. Two mistakes made while building the benchmark itself, and one wrong figure in a guide, were found and corrected on the way (section 5).

1. What was validated, and why this case

The course asks for the present worth of two alternatives to be calculated by hand and compared with the tool. We chose the two air-conditioning options in the demo because they differ in every feature the engine must handle: different lives (15 and 25 years), a re-purchase inside the study period, a separately wearing part, salvage value at year 25, different O&M and different savings. The benchmark is a plain Excel workbook, **Validation_Benchmark.xlsx**, built only from formulas and the input assumptions; it does not call the tool's code. The tool's results are typed in beside it for comparison.

2. The benchmark case

Project assumption	Value	Alternative	A: VRF	B: Chiller
Study period N	25 years	Capex (extra over code-minimum)	Rs 1,57,50,000	Rs 2,45,00,000
MARR (nominal)	10%	Life of the measure	15 years	25 years
General inflation	5% a year	O&M, year 1	Rs 1,75,000	Rs 9,00,000
Grid tariff, year 1	Rs 13 per kWh	Energy saved, year 1	2,48,000 kWh	3,46,000 kWh
Tariff escalation	5% a year	Degradation of savings	0.5% a year	0.5% a year
		Replaceable part	none	Rs 35,00,000 every 12 years

Source: the demo input workbook (all values are sourced estimates for a 12,000 m² Pune office). Costs and savings are incremental to the code-minimum option.

3. Method of the hand calculation

- **Year-by-year (sheets VRF and Chiller).** For each year 0 to 25: capex; O&M = $O\&M_1 \times 1.05^{t-1}$; energy saving = kWh × 0.995^{t-1} × Rs 13 × 1.05^{t-1}; re-purchases at inflated prices; salvage at year 25. Net cash flow × $1/(1.10)^t$ gives the present worth; the sum is the life-cycle cost.
- **Closed-form factors (sheet Factor check).** The same quantities by CE 725 Ch 3 formulas: $P/A_1(g, i, n) = (1 - ((1+g)/(1+i))^n) / (i - g)$ for the growing O&M and savings, and $P = F \times ((1+f)/(1+i))^t$ for each single re-purchase and for the residual.
- **Ratios (Results sheet).** EUAC = LCC × (A/P, 10%, 25) = LCC × 0.110168. BCR = PW of savings / (PW capex + O&M + re-purchases − residual). Discounted payback by linear interpolation of the cumulative present worth. Rate of return by Excel's MIRR with finance and reinvestment rate = MARR, because the cash flows change sign more than once (Ch 8).

- **Incremental analysis (sheet Incremental).** The gain of choosing the chiller over the VRF each year, its present worth and its MIRR (Ch 7).

4. Results: hand versus tool

Metric	A: VRF hand (Excel)	A: VRF tool	B: Chiller hand (Excel)	B: Chiller tool	Largest difference
PW of capex	15,750,000.00	15,750,000.00	24,500,000.00	24,500,000.00	0.00
PW of O&M	2,406,085.47	2,406,085.47	12,374,153.83	12,374,153.83	0.00
PW of re-purchases	7,838,442.32	7,838,442.32	3,148,758.80	3,148,758.80	0.00
PW of residual (salvage)	1,007,352.95	1,007,352.95	955,004.75	955,004.75	0.00
PW of energy savings	42,262,975.52	42,262,975.52	58,963,667.45	58,963,667.45	0.00
Life-cycle cost (present worth)	-17,275,800.68	-17,275,800.68	-19,895,759.58	-19,895,759.58	0.00
Cost per year (EUAC)	-1,903,241.66	-1,903,241.66	-2,191,877.48	-2,191,877.48	0.00
Benefit-cost ratio	1.691387	1.691387	1.509261	1.509261	0.00
Discounted payback (years)	6.5331	6.5331	9.2189	9.2189	0.00
Rate of return (MIRR at MARR)	12.5579%	12.5579%	12.6185%	12.6185%	0.00

Incremental (B over A)	Hand (Excel)	Tool	Difference
Extra capex of the chiller (Rs)	8,750,000.00	8,750,000.00	0
Present-worth gain of the chiller (Rs)	2,619,958.90	2,619,958.90	0
Incremental MIRR at the MARR	10.9061%	10.9061%	0
Sign changes in the gain series	5	5	0

Negative life-cycle cost means the measure pays for itself over 25 years. The tool and the benchmark agree that the chiller has the lower present worth (−Rs 1.99 Cr against −Rs 1.73 Cr), while the VRF has the lower upfront cost and the higher BCR. The extra Rs 87.5 L spent on the chiller earns 10.9% a year (MIRR), just above the 10% MARR.

Spot checks by calculator (independent of both):

Item	Hand calculation	Result
VRF re-purchase in year 15	Rs 1,57,50,000 × 1.05^{15}	Rs 3,27,43,119; present worth Rs 78,38,442
VRF saving, year 1	2,48,000 kWh × Rs 13	Rs 32,24,000
VRF salvage at year 25	Rs 3,27,43,119 × (15 + 15 − 25)/15	Rs 1,09,14,373; present worth Rs 10,07,353
Chiller part overhauls	35,00,000 × 1.05^{12} and 35,00,000 × 1.05^{24}	Rs 62,85,497 (year 12), Rs 1,12,87,850 (year 24)
Chiller part salvage at year 25	35,00,000 × 1.05^{24} × 11/12	Rs 1,03,47,196; present worth Rs 9,55,005

5. Discrepancies and mistakes found, and how each was resolved

The final comparison shows no numerical difference. This table records what was found on the way, so the validation is honest about the work behind it.

No.	Found by	What was found	Cause	Resolution
1	Benchmark verification (formulas evaluated independently)	The VRF re-purchase column returned a divide-by-zero error.	AI-written formula tested a replaceable part's timing even where the measure has no part (life 0).	Formula rewritten to test the part only if it exists. Re-verified.
2	Benchmark verification	Life-cycle cost of the benchmark was far from the tool's value (Rs 41 L instead of −Rs 1.99 Cr for the chiller).	AI-written total row discounted the present-worth column a second time.	Total row changed to a plain sum. All 14 comparisons then matched.
3	Benchmark verification	A count of sign changes for the incremental cash flow was evaluated as 0; the sheet's data has 5.	An array formula was not evaluated reliably.	Replaced by a helper column. Now 5, equal to the tool.
4	Cross-check of guide against inputs	How-it-works guide said LED lighting degrades 0.5% a year; the data says 1%.	AI-written explanatory text, not the tool; net savings growth of 3.95% in the same table already used 1%.	Guide corrected to 1% and rebuilt.
5	Textbook examples as tests	Two worked examples in the CE 725 notes differ slightly from exact calculation (Ch 5 PW_A 57,875.3 vs 57,877.20; Ch 8 IRRs 23.27% / 34.26% vs 23.40% / 34.36%).	Rounding in the notes.	Tests use the exact values; the differences are recorded here.
6	Sanity check of first demo data	Early demo energy savings were too high: every measure paid back and nothing flipped.	AI-generated placeholder values.	Demo data rebuilt from one sizing basis with sources (values still need verification against real tenders).

6. Edge cases and automatic tests

Beyond the benchmark, 82 automatic tests run after every change:

Test file	Tests	What it checks
test_core.py	26	Engine hand cases (annuity, re-purchase, residual, escalation, degradation, solar self-use and export), IGBC scoring, optimizer, risk, Monte Carlo
test_course.py	14	CE 725 worked examples: interest factors, present worth, EUAW, IRR, multiple IRR, MIRR
test_edge_cases.py	13	Bad or missing input rows, wrong file type, blank groups, zero-benefit measure, empty library, too many combinations, 5-year study period, report with no feasible mix, Excel export agrees with engine
test_exports.py	12	PDF report content, results workbook, input workbook round trip
test_ui.py	16	Every screen loads with and without a project
test_launcher.py	1	Launcher and app files compile

7. Limits of this validation

- The hand benchmark covers one pair of alternatives in full; the other measures are covered by the automatic tests and by the same engine code path.
- The benchmark checks that the tool computes its stated method correctly. It does not validate the demo's prices, tariffs or IGBC point tables, which remain sourced estimates and illustrative values.
- The benchmark and the tool share conventions chosen by the team (nominal rupees, planning-horizon method, linear residual). Different conventions would give different, equally valid numbers.